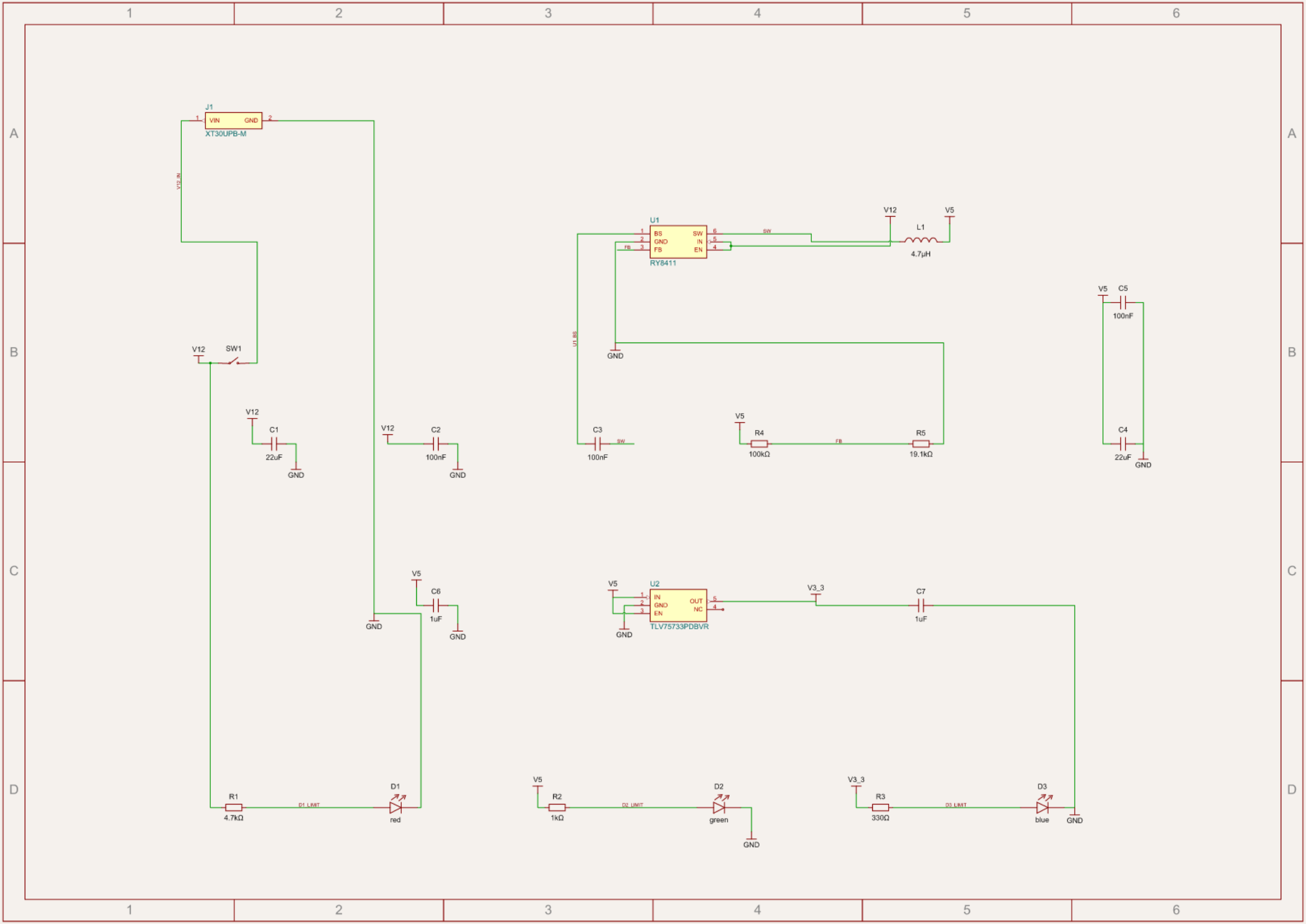
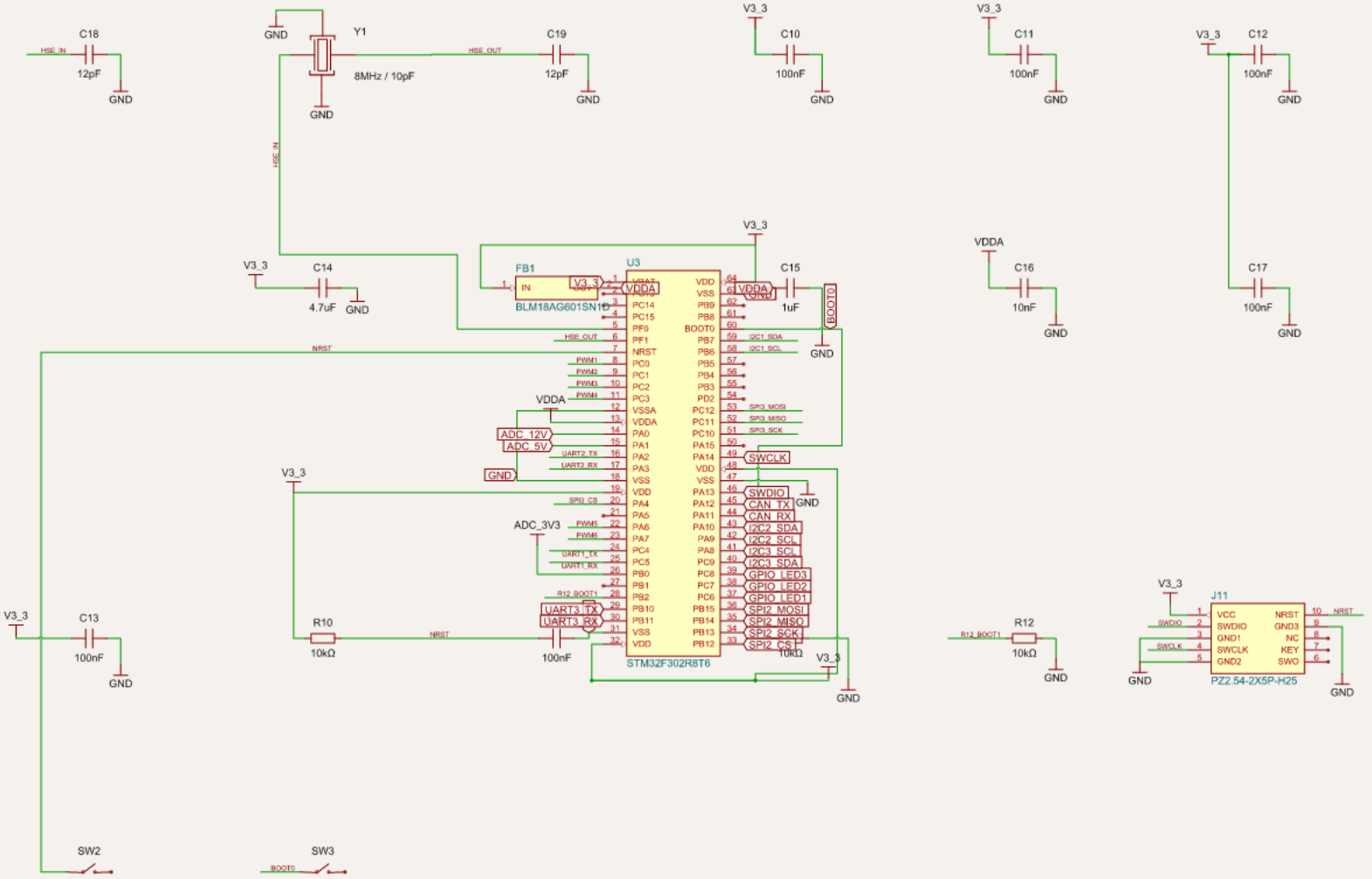
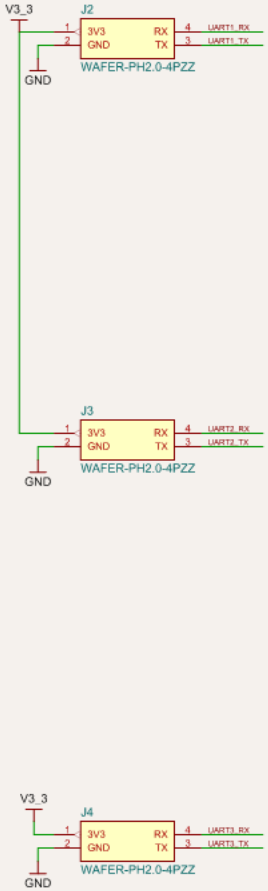
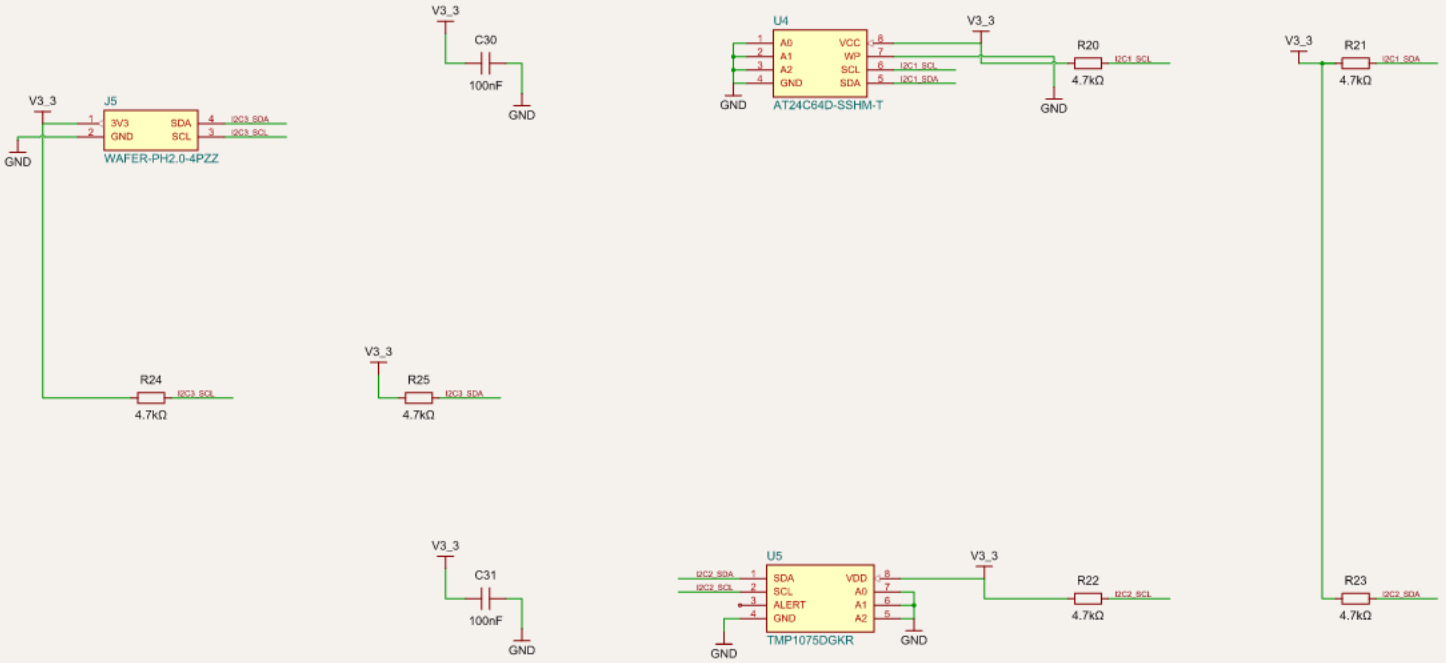
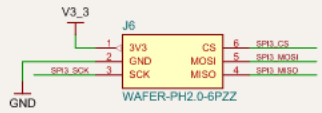

J1
VIN
GND
XT30UPB-M
V12_IN
V12
SW1
V12
C1
22uF
GND
V12
C2
100nF
GND
V5
C6
1uF
GND
GND
U1
BS
GND
FB
SW
IN
EN
RY8411
SW
V12
L1
V5
4.7µH
U1_BS
GND
C3
100nF
SW
V5
R4
100kΩ
FB
R5
19.1kΩ
V5
C5
100nF
C4
22uF
GND
V5
U2
IN
GND
EN
OUT
NC
TLV75733PDBVR
GND
V3_3
C7
1uF
R1
4.7kΩ
D1_LIMIT
D1
red
V5
R2
1kΩ
D2_LIMIT
D2
green
GND
V3_3
R3
330Ω
D3_LIMIT
D3
blue
GND

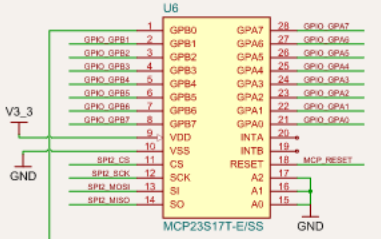

C18
12pF
HSE_IN
GND
Y1
8MHz / 10pF
HSE_OUT
C19
12pF
C10
100nF
C11
100nF
C12
100nF
V3_3
C14
4.7uF
FB1
BLM18AG601SN1D
U3
STM32F302R8T6
C15
1uF
VDDA
C16
10nF
C17
100nF
BOOT0
NRST
SWCLK
SWDIO
CAN_TX
CAN_RX
I2C2_SDA
I2C2_SCL
I2C3_SCL
I2C3_SDA
GPIO_LED3
GPIO_LED2
GPIO_LED1
SPI2_MOSI
SPI2_MISO
SPI2_SCK
SPI2_CS
ADC_12V
ADC_5V
ADC_3V3
UART3_TX
UART3_RX
C13
100nF
R10
10kΩ
R12
10kΩ
J11
PZ2.54-2X5P-H25
SW2
SW3


V3_3
J2
1
2
3V3
GND
RX
TX
4
3
UART1_RX
UART1_TX
GND
WAFER-PH2.0-4PZZ
J3
1
2
3V3
GND
RX
TX
4
3
UART2_RX
UART2_TX
GND
WAFER-PH2.0-4PZZ
V3_3
J4
1
2
3V3
GND
RX
TX
4
3
UART3_RX
UART3_TX
GND
WAFER-PH2.0-4PZZ

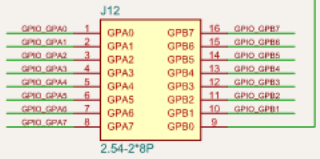

V3_3
J5
3V3
GND
SDA
SCL
I2C3_SDA
I2C3_SCL
WAFER-PH2.0-4PZZ
GND
C30
100nF
GND
U4
A0
A1
A2
GND
VCC
WP
SCL
SDA
I2C1_SCL
I2C1_SDA
AT24C64D-SSHM-T
GND
V3_3
R20
4.7kΩ
I2C1_SCL
GND
V3_3
R21
4.7kΩ
I2C1_SDA
R24
4.7kΩ
I2C3_SCL
V3_3
R25
4.7kΩ
I2C3_SDA
V3_3
C31
100nF
GND
U5
I2C2_SDA
I2C2_SCL
SDA
SCL
ALERT
GND
VDD
A0
A1
A2
TMP1075DGKR
GND
V3_3
R22
4.7kΩ
I2C2_SCL
R23
4.7kΩ
I2C2_SDA

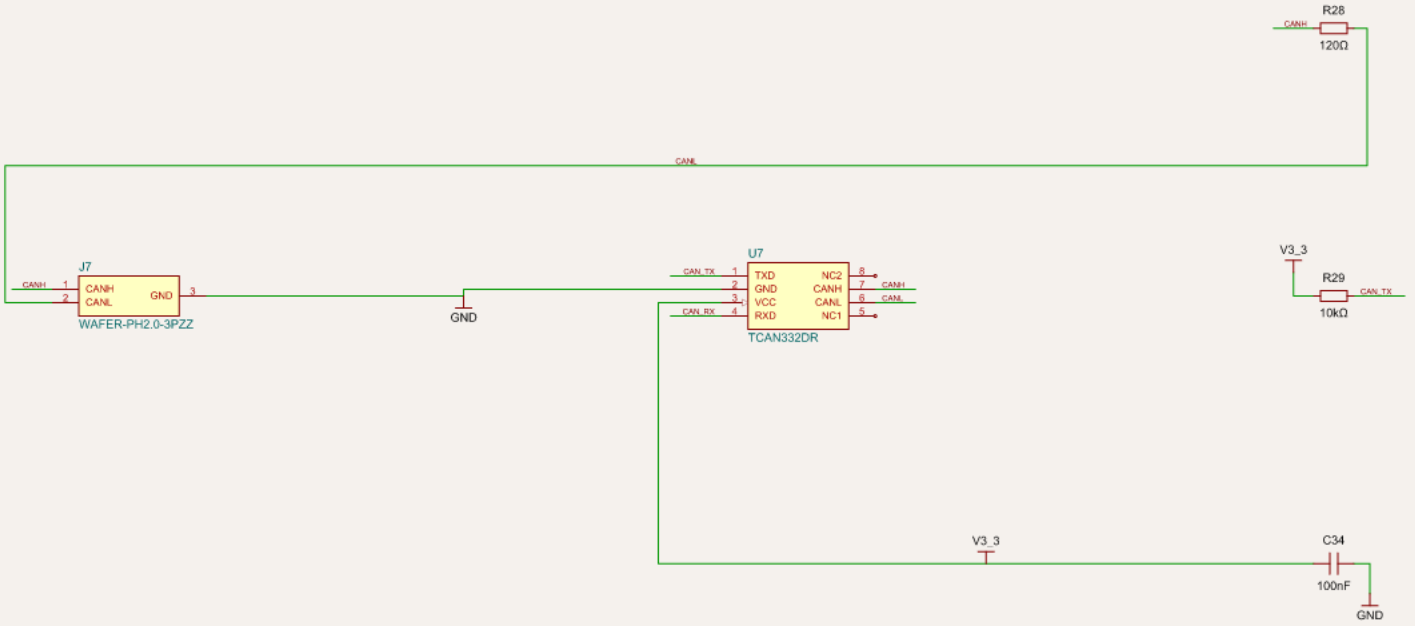

V3_3
J6
3V3
GND
SCK
CS
MOSI
MISO
SPI3_CS
SPI3_MOSI
SPI3_MISO
SPI3_SCK
GND
WAFER-PH2.0-6PZZ


R26
MCP_RESET
10kΩ
V3_3
R27
SPI2_CS
10kΩ

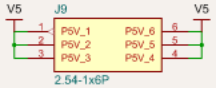

U6
GPB0
GPB1
GPB2
GPB3
GPB4
GPB5
GPB6
GPB7
VDD
VSS
CS
SCK
SI
SO
GPA7
GPA6
GPA5
GPA4
GPA3
GPA2
GPA1
GPA0
INTA
INTB
RESET
A2
A1
A0
GPIO_GPB1
GPIO_GPB2
GPIO_GPB3
GPIO_GPB4
GPIO_GPB5
GPIO_GPB6
GPIO_GPB7
SPI2_CS
SPI2_SCK
SPI2_MOSI
SPI2_MISO
GPIO_GPA7
GPIO_GPA6
GPIO_GPA5
GPIO_GPA4
GPIO_GPA3
GPIO_GPA2
GPIO_GPA1
GPIO_GPA0
MCP_RESET
V3_3
GND
GND
MCP23S17T-E/SS

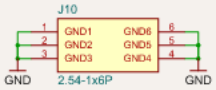

V3_3
C32
100nF
GND
V3_3
C33
100nF
GND

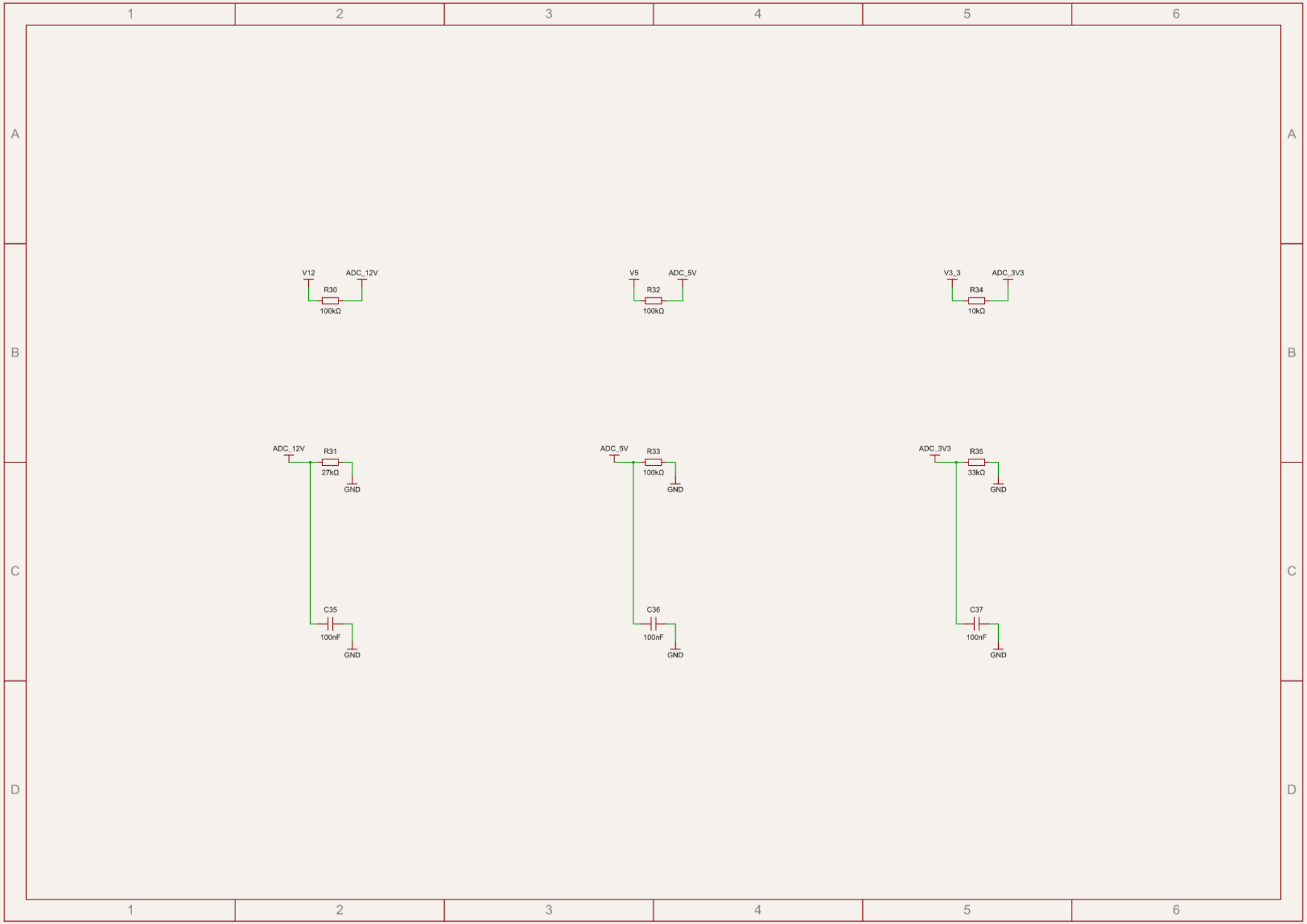

J12
GPA0
GPA1
GPA2
GPA3
GPA4
GPA5
GPA6
GPA7
GPB7
GPB6
GPB5
GPB4
GPB3
GPB2
GPB1
GPB0
GPIO_GPA0
GPIO_GPA1
GPIO_GPA2
GPIO_GPA3
GPIO_GPA4
GPIO_GPA5
GPIO_GPA6
GPIO_GPA7
GPIO_GPB7
GPIO_GPB6
GPIO_GPB5
GPIO_GPB4
GPIO_GPB3
GPIO_GPB2
GPIO_GPB1
GPIO_GPB0
2.54-2*8P

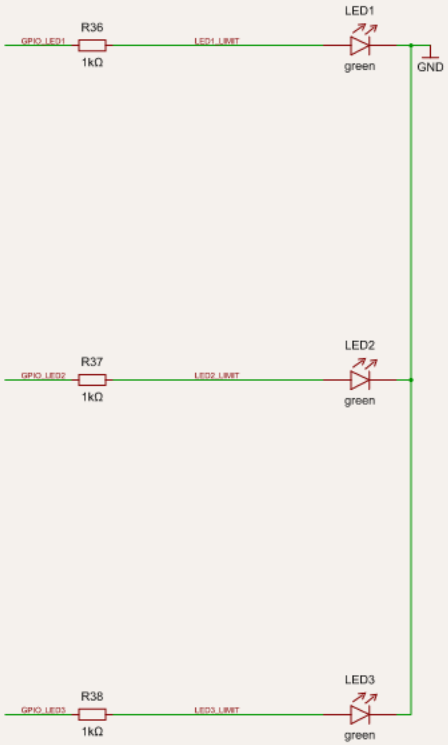

R28
CANH
120Ω
CANL
J7
CANH
CANL
GND
WAFER-PH2.0-3PZZ
GND
U7
CAN_TX
CAN_RX
TXD
GND
VCC
RXD
NC2
CANH
CANL
NC1
TCAN332DR
V3_3
R29
CAN_TX
10kΩ
V3_3
C34
100nF
GND


J8
PWM1
PWM2
PWM3
PWM6
PWM5
PWM4
2.54-1x6P


V5
J9
V5
P5V_1
P5V_2
P5V_3
P5V_6
P5V_5
P5V_4
2.54-1x6P


J10
GND1
GND2
GND3
GND6
GND5
GND4
GND
2.54-1x6P
GND


1
2
3
4
5
6
A
B
C
D
V12
ADC_12V
R30
100kΩ
V5
ADC_5V
R32
100kΩ
V3_3
ADC_3V3
R34
10kΩ
ADC_12V
R31
27kΩ
GND
C35
100nF
GND
ADC_5V
R33
100kΩ
GND
C36
100nF
GND
ADC_3V3
R35
33kΩ
GND
C37
100nF
GND


R36
GPIO_LED1
1kΩ
LED1_LIMIT
LED1
green
GND
R37
GPIO_LED2
1kΩ
LED2_LIMIT
LED2
green
R38
GPIO_LED3
1kΩ
LED3_LIMIT
LED3
green